

Subject Description Form

Subject Code	COMP5327
Subject Title	Wireless Networking and Mobile Computing
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil (but some knowledge in internet infrastructure and protocols is preferable)
Objectives	After completing this subject, students will learn about: • enabling technologies for wireless networking and mobile computing • wireless networking standards • mobile computing applications
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. show in-depth understanding of advanced wireless networking technologies and their applications in a variety of real-world applications, addressing complex interdisciplinary issues with professionalism; b. work with apply coherent and state-of-the-art knowledge on related protocols and technologies, design innovative solutions and critically assess their performance, leveraging emerging technologies for global challenges; c. categorize and summarize the trends and development of wireless networking and mobile computing, continuously adapting and reflecting on personal growth. d. demonstrate team work spirit in group projects, communicate appropriately with a range of audiences in presentation and summarize results in technical reports, fostering intercultural and interdisciplinary interactions.
Subject Synopsis/ Indicative Syllabus	1) Cellular Networks and 5G Frequency Reuse, Access Protocols, Location Management, Handoff Management, Cellular Network Standards (2G/3G/4G/5G), Key Features and Advancements in 5G (e.g., enhanced mobile broadband, ultra-reliable low-latency communication, massive IoT connectivity) 2) Wireless Local Area Networks Overview of IEEE 802.11 Standards (Wi-Fi 4/5/6), Access Protocols, Mobility Management, Advances in High-Efficiency WLANs 3) Personal Area Networks and Bluetooth Overview of Bluetooth Standards, Piconet and Scatternet, Frequency Hopping, Baseband Protocol, Link Manager Protocol, Logical Link Control and Adaptation Protocol, Bluetooth Low Energy (BLE) 4) Mobile/Wireless Security Security in Cellular Networks (including 5G), Wi-Fi Security, Bluetooth Security, Security Challenges in Satellite Communication

	5) Location-aware Computing GPS, Indoor Positioning Techniques, Location-aware Applications, Emerging Positioning Solutions in 5G and Satellite Networks 6) Mobile Computing Applications Mobile Apps, Edge Computing for Mobile Devices, Case Studies of Real-world Applications 7) LEO Satellite Communication Fundamentals of Low Earth Orbit (LEO) Satellite Networks, Architecture and Communication Protocols, Integration with Terrestrial Networks, Applications in Global Connectivity and IoT 8) Other Selected Topics Mobile IP, RFID, Wireless Sensor Networks, Recent Trends and Emerging Technologies in Mobile and Wireless Communication																																					
Teaching/Learning Methodology	Teaching is mainly conducted through lectures. Learning is supplemented by labs/tutorials, seminars and e-learning methods, where applicable. Students are assessed through assignments, a project and an examination.																																					
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>1. Assignments</td><td rowspan="2">30</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>2. Project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>3. Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>Total</td><td>100</td><td colspan="4"></td></tr> </tbody> </table> The project is used to assess all learning outcomes (a) – (d). The assignments are used as a continuous assessment method to assess learning outcomes (a) – (c) (e.g., students’ understanding of the technologies). Finally, students are assessed by a formal examination, covering learning outcomes (a) – (c).					Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Assignments	30	✓	✓	✓		2. Project	✓	✓	✓	✓	3. Examination	70	✓	✓	✓		Total	100				
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1. Assignments	30	✓	✓	✓																																		
2. Project		✓	✓	✓	✓																																	
3. Examination	70	✓	✓	✓																																		
Total	100																																					
Student Study Effort Expected	Class contact:																																					
	▪ Class activities (lectures/tutorials/lab(s))		39 Hrs.																																			
	Other student study effort:																																					
	▪ Self-study, assignments, project, exam		66 Hrs.																																			
	Total student study effort		105 Hrs.																																			

Reading List and References	Books: [1]. W. Stallings, <i>Wireless Communications & Networks</i>, 2nd ed., NJ: Pearson Education, 2011 [2]. Raj Kamal, <i>Mobile Computing</i>, 3rd ed., Oxford, UK: Oxford University Press, 2018. [3]. X. Lin and N. Zein, <i>5G and Beyond: Fundamentals and Standards</i>. Cham: Springer, 2021. Journals: IEEE Transactions on Mobile Computing IEEE Pervasive Computing IEEE Transactions on Wireless Communications IEEE Journal on Selected Areas in Communications ACM Wireless Networks ACM Mobile Networks and Applications
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